Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 7

Time Allowed: 30 Minutes

General Instructions:

Read the following instructions carefully.

1. There are 16 questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

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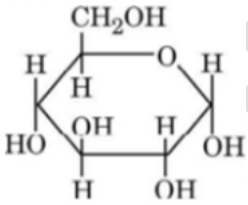
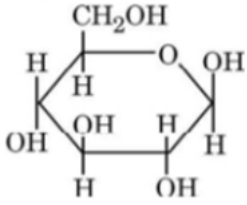
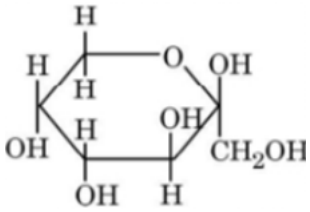
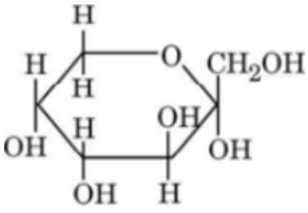
Section A

1. Butanenitrile may be prepared by heating: **[1]**

   a) propyl chloride with KCN
   b) propyl alcohol with KCN
   c) butyl alcohol with KCN
   d) butyl chloride with KCN

2. Which of the following structures represents α-D-glucose? **[1]**

   a)
   
   b)
   
   c)
   
   d)
   

3. The formation of cyanohydrin from an aldehyde is an example of: **[1]**

   a) nucleophilic substitution
   b) electrophilic substitution
   c) nucleophilic addition
   d) electrophilic addition

4. Consider the following reaction: **[1]**

$$\begin{matrix} H \\ H \end{matrix}\!\!>C=O + \begin{matrix} H \\ H \end{matrix}\!\!>C=O + \text{Conc} \cdot KOH \xrightarrow{\Delta} A + B$$

Identify A and B from the given options:

a) A - Methanol, B - Potassium acetate
b) A - Ethanol, B - Potassium formate
c) A - Methanol, B - Potassium formate
d) A - Methanal, B - Ethanol

5. The half-life of a reaction is halved as the initial concentration of the reactant is doubled. The order of the reaction is: **[1]**

a) 3
b) 1
c) 2
d) 0

6. Match the items given in column I with that in column II: **[1]**

Column I	Column II
(a) The cell reaction $2H_2 + O_2 \rightarrow 2H_2O$.	(i) Nickel-cadmium Storage Cell.
(b) Anodic Reaction is $Zn \rightarrow Zn^{2+} + 2e^-$.	(ii) Mercury Cell.
(c) Cathodic Reaction $HgO(s) + H_2O + 2e^- \rightarrow Hg(l) + 2OH^-$.	(iii) Bacon Cell.
(d) Anodic Reaction $Cd(s) + 2OH^-(aq) \rightarrow Cd(OH)_2(s) + 2e^-$.	(iv) Leclanche cell.

a) (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)
b) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
c) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
d) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

7. Which one of the following compounds is more reactive towards S_N1 reaction? **[1]**

a) $C_6H_5CH_2Br$
b) $C_6H_5CH(CH_3)Br$
c) $C_6H_5CH(C_6H_5)Br$
d) $CH_2 = CHCH_2Br$

8. Deficiency of which of the following vitamins causes Pernicious anaemia? **[1]**

a) Vitamin B_2
b) Vitamin B_6
c) Vitamin B_1
d) Vitamin B_{12}

9. The expression which gives 3/4th life of the first-order reaction is: **[1]**

a) $\frac{k}{2.303}\log 4/3$
b) $\frac{2.303}{k}\log 4$
c) $\frac{2.303}{k}\log 3$
d) $\frac{k}{2.303}\log 3/4$

10. Which of the following compounds will give butanone on oxidation with alkaline $KMnO_4$ solution? **[1]**

a) Butan-2-ol
b) Butan-1-ol
c) Butan-3-ol
d) Both Butan-2-ol and Butan-1-ol

11. Which of the following is most acidic? **[1]**

a) Benzyl alcohol
b) m – Chlorophenol
c) Phenol
d) Cyclohexanol

12. In the reaction **[1]**

$C_6H_5NH_2 + CHCl_3 + 3\ KOH \rightarrow A + 3B + 3C$ the product A is

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a) C_6H_5CN

b) C_6H_5NC

[1]

c) $C_6H_5NHCH_3$

d) C_6H_5Cl

13. **Assertion (A):** Oils float on water.

Reason (R): Oils are the naturally occurring compounds that belong to lipid.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion (A):** Nucleophilic substitution of iodoethane is easier than chloroethane. **[1]**

Reason (R): Bond enthalpy of C-I bond is less than that of C-Cl bond.

a) Both (A) and (R) are true and (R) is the correct explanation of (A).

b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

c) (A) is true, but (R) is false.

d) (A) is false, but (R) is true.

15. **Assertion (A):** Nucleophilic substitution of iodoethane is easier than chloroethane. **[1]**

Reason (R): Bond energy of C - Cl bond is less than C - I bond.

a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

c) Assertion (A) is true, but Reason (R) is false.

d) Assertion (A) is false, but Reason (R) is true.

16. **Assertion (A):** p-nitrophenol is more acidic than phenol. **[1]**

Reason (R): Nitro group helps in the stabilization of the phenoxide ion by dispersal of negative charge due to resonance.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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